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Sotatercept in Patients with Pulmonary Arterial Hypertension at High Risk for Death

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ABSTRACT

BACKGROUND

Sotatercept improves exercise capacity and delays the time to clinical worsening in patients with World Health Organization (WHO) functional class II or III pulmonary arterial hypertension. The effects of add-on sotatercept in patients with advanced pulmonary arterial hypertension and a high risk of death are unclear.

METHODS

In this phase 3 trial, we randomly assigned patients with pulmonary arterial hypertension (WHO functional class III or IV) and a high 1-year risk of death (Registry to Evaluate Early and Long-Term Pulmonary Arterial Hypertension Disease Management Lite 2 risk score, ≥9) who were receiving the maximum tolerated dose of background therapy to receive add-on sotatercept (starting dose, 0.3 mg per kilogram of body weight; escalated to target dose, 0.7 mg per kilogram) or placebo every 3 weeks. The primary end point was a composite of death from any cause, lung transplantation, or hospitalization (≥24 hours) for worsening pulmonary arterial hypertension, assessed in a time-to-first-event analysis.

RESULTS

A total of 172 patients were included (86 each in the sotatercept and placebo groups). The trial was stopped early on the basis of the efficacy results of a prespecified interim analysis. At least one primary end-point event occurred in 15 patients (17.4%) in the sotatercept group and in 47 patients (54.7%) in the placebo group (hazard ratio, 0.24; 95% confidence interval, 0.13 to 0.43; P<0.001). Death from any cause occurred in 7 patients (8.1%) in the sotatercept group and in 13 patients (15.1%) in the placebo group; lung transplantation in 1 patient (1.2%) and 6 patients (7.0%), respectively; and hospitalization for worsening pulmonary arterial hypertension in 8 patients (9.3%) and 43 patients (50.0%). The most common adverse events with sotatercept were epistaxis and telangiectasia.

CONCLUSIONS

Among high-risk adults with pulmonary arterial hypertension who were receiving the maximum tolerated dose of background therapy, treatment with sotatercept resulted in a lower risk of a composite of death from any cause, lung transplantation, or hospitalization (≥24 hours) for worsening pulmonary arterial hypertension than placebo. (Funded by Merck Sharp and Dohme, a subsidiary of Merck [Rahway, NJ]; ZENITH ClinicalTrials.gov number, NCT04896008.)

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*A complete list of the ZENITH trial investigators is provided in the Supplementary Appendix, available at NEJM.org.

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Pulmonary arterial hypertension is a progressive, potentially fatal disease characterized by remodeling of the pulmonary vasculature, which results in increased pulmonary vascular resistance and pulmonary artery pressure and can ultimately lead to right ventricular failure and death.[1,2] Established treatments for pulmonary arterial hypertension include endothelin-receptor antagonists, phosphodiesterase-5 inhibitors, soluble guanylate cyclase stimulators, and prostacyclin-pathway agents.[3] Although randomized, placebo-controlled trials of existing therapies have shown delays in the time to clinical worsening, these findings primarily reflected functional deterioration rather than major events such as death, lung transplantation, and hospitalization for worsening pulmonary arterial hypertension.[4-7]

Sotatercept, a first-in-class activin-signaling inhibitor, offers a novel therapeutic alternative to vasodilators in the treatment of pulmonary arterial hypertension[8-10] by binding proproliferative members of the transforming growth factor β superfamily (e.g., activins), targeting pulmonary vascular remodeling.[11-13] In the phase 3 STELLAR trial, which involved patients with group 1 pulmonary arterial hypertension in World Health Organization (WHO) functional class II or III (classes range from I to IV, with higher numbers indicating greater functional limitations), the addition of sotatercept to background therapy improved the 6-minute walk distance, pulmonary vascular resistance, and N-terminal pro–B-type natriuretic peptide (NT-proBNP) levels, while reducing the risk of clinical worsening events by 84%.[8]

In the ZENITH trial, we assessed the effect of sotatercept on major outcomes in patients with more advanced pulmonary arterial hypertension (WHO functional class III or IV) and a risk score of 9 or higher on the REVEAL (Registry to Evaluate Early and Long-Term Pulmonary Arterial Hypertension Disease Management) Lite 2 risk calculator (a multiparametric prognostic tool; scores range from 1 to 14, with higher scores indicating a greater 1-year risk of death[14,15]) who were receiving maximum tolerated doses of double or triple therapy for pulmonary arterial hypertension.

METHODS

TRIAL DESIGN AND OVERSIGHT

ZENITH was a phase 3, multicenter, double-blind, randomized, placebo-controlled trial in which the efficacy and safety of sotatercept was evaluated in patients with pulmonary arterial hypertension at high risk for death (Fig. S1 in the Supplementary Appendix, available with the full text of this article at NEJM.org). Patients were randomly assigned in a 1:1 ratio to receive add-on subcutaneous sotatercept (starting dose, 0.3 mg per kilogram of body weight; escalated to target dose, 0.7 mg per kilogram) or placebo every 21 days, in addition to stable maximum tolerated doses of double or triple therapy for pulmonary arterial hypertension. Randomization was stratified according to the REVEAL Lite 2 risk score (9 or 10 vs. ≥11) and pulmonary arterial hypertension subtype (associated with connective-tissue disease vs. not associated with connective-tissue disease). Dose adjustments followed prespecified criteria that were based on hemoglobin levels and platelet counts, with dose modifications indicated for the occurrence of telangiectasia and serious bleeding.

The original protocol specified that all the patients were to remain in the trial until the planned number of primary end-point events had occurred. However, after the results of the STELLAR trial[8] became available, a protocol amendment was implemented to allow patients who had been hospitalized for at least 24 hours for worsening pulmonary arterial hypertension to transition into the open-label long-term follow-up study SOTERIA (ClinicalTrials.gov number, NCT04796337).

The ZENITH trial was conducted in accordance with the principles of Good Clinical Practice originating from the Declaration of Helsinki. Written informed consent was obtained from all the patients before enrollment. The protocol (available at NEJM.org) and all amendments were approved by the appropriate ethics body at each participating site. An independent data monitoring committee provided oversight and had the authority to recommend trial modifications or termination.

The sponsor (Merck Sharp and Dohme, a subsidiary of Merck [Rahway, NJ]) and a steering committee designed the trial. The sponsor oversaw the monitoring of the trial and performed the analyses. The investigators worked under confidentiality agreements with the sponsor. All the authors participated in interpretation of the data, review of the manuscript, and the decision to submit the manuscript for publication. The academic authors had full access to all

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the data. All the authors had the authority to request ad hoc analyses. The first author drafted the initial manuscript with editorial support from a sponsor-employed medical writer. All the authors vouch for the completeness and accuracy of the data and for the fidelity of the trial to the protocol.

TRIAL POPULATION

Eligible patients were adults (18 to 75 years of age) who had symptomatic WHO group 1 pulmonary arterial hypertension (idiopathic, heritable, drug- or toxin-induced, connective-tissue disease–associated, or with simple congenital systemic-to-pulmonary shunts ≥1 year after repair) in WHO functional class III or IV and had a REVEAL Lite 2 risk score of 9 or higher. Hemodynamic criteria assessed during the screening period (i.e., within 6 weeks before randomization) included a pulmonary vascular resistance of at least 400 dyn·s·cm^{-5} (≥5 Wood units) and a pulmonary artery wedge pressure or left ventricular end-diastolic pressure of 15 mm Hg or lower. Patients were receiving stable maximum tolerated doses of double or triple therapy for pulmonary arterial hypertension beginning at least 30 days before screening. Women used highly effective contraception; men used barrier methods.

Key exclusion criteria were pulmonary arterial hypertension associated with portal hypertension, human immunodeficiency virus infection, pulmonary veno-occlusive disease, or pulmonary capillary hemangiomatosis; pulmonary arterial hypertension other than group 1; a left ventricular ejection fraction below 45%; clinically significant mitral or aortic valve disease; and other clinically significant coexisting conditions or laboratory abnormalities (e.g., a hemoglobin level above sex-specific upper limits of the normal range or a platelet count of <50,000 per mm^3).

EFFICACY END POINTS

The primary end point was a composite of death from any cause, lung transplantation, or hospitalization lasting at least 24 hours for worsening pulmonary arterial hypertension, assessed in a time-to-first-event analysis. The secondary end points were overall survival and transplantation-free survival, assessed in time-to-event analyses; death from any cause; the change from baseline in the REVEAL Lite 2 risk score at week 24[14,15]; a low or intermediate REVEAL Lite 2 risk score (≤7) at week 24; the changes from baseline in the NT-proBNP level, the mean pulmonary artery pressure, and pulmonary vascular resistance at week 24; improvement in WHO functional class at the end of the double-blind, placebo-controlled treatment period; and changes from baseline in the 6-minute walk distance, cardiac output, and the index score on the EuroQol 5-Dimension 5-Level (EQ-5D-5L) questionnaire (the total score of five dimensions of health-related quality of life, weighted according to country-specific or region-specific preferences; scores range from <0 to 1, with higher scores indicating better health-related quality of life) at week 24.

All potential primary end-point events were adjudicated by an independent adjudication committee whose members were unaware of the trial-group assignments, with one exception: deaths that occurred after withdrawal from the trial in patients who had had no primary end-point event during the trial were not adjudicated. Lung transplantation and hospitalization (≥24 hours) for worsening pulmonary arterial hypertension were recorded only during the ZENITH trial, whereas deaths were recorded both during and after the trial, up to the data cutoff for the interim analysis.

An interim analysis was planned to be performed after approximately 59 events had accrued. For the safety assessments, patients were followed from day 1 (the start of administration of sotatercept or placebo) through the last dose date plus 56 days.

SAFETY END POINTS

Safety assessments included adverse events, laboratory test results, vital signs, and physical examination findings. Adverse events of interest (increased hemoglobin level, thrombocytopenia, immunogenicity, increased blood pressure or hypertension, thromboembolic events, bleeding events, renal toxic effects, hepatic toxic effects, and cardiac events) and an adverse event of special interest (telangiectasia) were identified with the use of standardized *Medical Dictionary for Regulatory Activities* (MedDRA) queries or grouped preferred terms that were selected on the basis of the mechanism of action of sotatercept, preclinical data, or previous clinical findings.

STATISTICAL ANALYSIS

The sample size was determined with the use of East software, version 6.4, and was based on the primary efficacy end point. Assuming a hazard ratio of 0.55 and a planned interim analysis after

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the occurrence of approximately 50% of the anticipated primary efficacy end-point events, it was estimated that 118 events would need to occur by the time of the final analysis to provide the trial with 90% power (with a one-sided type I error rate of 0.025). A total of approximately 166 patients were planned to be enrolled.

All the efficacy analyses were performed in the intention-to-treat population, which included all the patients who had undergone randomization, irrespective of whether they received any trial medication. Safety analyses were performed in the safety population, which comprised all the patients who had undergone randomization and had received at least one dose of sotatercept or placebo.

The interim analysis was performed with the use of the Lan–Demets spending function (O'Brien–Fleming approximation) with a P-value boundary of 0.0021 (one-sided) or 0.0042 (two-sided). The type I error rate was controlled with the use of a hierarchical testing strategy, starting with the primary end point and continuing with the secondary end points in the order presented above, stopping at the first nonsignificant result. For end points that were ineligible for statistical testing, unadjusted 95% confidence intervals were provided. These confidence intervals should not be interpreted as definitive evidence of an effect in the absence of statistical testing.

The interim analysis included all available data up to the data-cutoff date of July 26, 2024. Time-to-event analyses for the primary efficacy end point and for other time-to-event end points were performed with the use of the log-rank test and Cox regression, stratified according to the REVEAL Lite 2 risk score (9 or 10 vs. ≥11) and pulmonary arterial hypertension subtype (associated with connective-tissue disease vs. not associated with connective-tissue disease). For time-to-event end points, missing follow-up data from patients who withdrew from the trial prematurely were censored at the date of withdrawal, and the analysis was performed under the missing-at-random assumption. Continuous end points were analyzed by means of the aligned-rank Wilcoxon test with the use of Hodges–Lehmann estimates for between-group differences. For patients with missing data owing to death, the worst-rank score was imputed; for patients with missing data after lung transplantation or hospitalization (≥24 hours) for worsening pulmonary arterial hypertension, the next worst-rank score was imputed. Other missing data were imputed by multiple imputation, under the assumption that the data were missing at random. Dichotomous variables were assessed with the use of the Cochran–Mantel–Haenszel test, stratified according to randomization factors; patients with missing data were considered as not having had a response. Details of the trial design and statistical methods, including sensitivity analyses, are provided in the Supplementary Appendix.

RESULTS

TRIAL POPULATION

The first patient was enrolled on December 1, 2021, and the primary completion date was July 26, 2024 (cutoff date of the interim analysis). During the screening period, 255 patients were assessed for eligibility, 82 (32.2%) of whom were excluded (Fig. S2). The primary reasons for screening failure were not having a REVEAL Lite 2 risk score of 9 or higher and not meeting the hemodynamic criteria (Table S1). A total of 173 patients underwent randomization, including 1 patient who did not meet the eligibility criteria and underwent randomization in error; the patient was immediately withdrawn from the trial without having received sotatercept or placebo and was excluded from all analyses. Consequently, 172 patients (86 in the sotatercept group and 86 in the placebo group) were included in the intention-to-treat population. The median duration of follow-up was 10.6 months (range, 0.3 to 26.1) in the sotatercept group and 7.1 months (range, 0.7 to 24.2) in the placebo group (Table S3).

The demographic and clinical characteristics of the patients at baseline were generally balanced, with more women and a higher percentage of patients with a body-mass index (the weight in kilograms divided by the square of the height in meters) of 30 or higher in the placebo group (Table 1 and Table S4). Table S6 shows the representativeness of the trial population. During the trial, adjustments to background therapy for pulmonary arterial hypertension were made at the discretion of the investigators on the basis of clinical judgement and were not mandated by the protocol (Table S7).

PRIMARY EFFICACY END POINT

On the basis of the results of the interim analysis of the primary end point, the data monitoring

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Table 1. Demographic and Baseline Clinical Characteristics.*

Characteristic	Sotatercept (N=86)	Placebo (N=86)	Total (N=172)
Female sex — no. (%)	61 (70.9)	71 (82.6)	132 (76.7)
Age — yr	55.3±14.3	53.5±14.3	54.4±14.3
Race — no. (%)†			
White	73 (84.9)	76 (88.4)	149 (86.6)
Other‡	12 (14.0)	10 (11.6)	22 (12.8)
Missing	1 (1.2)	0	1 (0.6)
Body-mass index ≥30 — no. (%)§	14 (16.3)	19 (22.1)	33 (19.2)
Time since diagnosis of pulmonary arterial hypertension — yr¶	7.2±5.6	8.2±6.7	7.7±6.2
Classification of pulmonary arterial hypertension — no. (%)			
Idiopathic	42 (48.8)	44 (51.2)	86 (50.0)
Heritable	11 (12.8)	7 (8.1)	18 (10.5)
Associated with connective-tissue disease	22 (25.6)	26 (30.2)	48 (27.9)
Drug-induced or toxin-induced	6 (7.0)	5 (5.8)	11 (6.4)
Associated with corrected congenital shunts	5 (5.8)	4 (4.7)	9 (5.2)
REVEAL Lite 2 risk score — no. (%)‖			
8–10	60 (69.8)	60 (69.8)	120 (69.8)
≥11	26 (30.2)	26 (30.2)	52 (30.2)
WHO functional class — no. (%)**			
III	66 (76.7)	62 (72.1)	128 (74.4)
IV	20 (23.3)	24 (27.9)	44 (25.6)
Background therapy for pulmonary arterial hypertension — no. (%)††			
Prostacyclin infusion therapy‡‡	53 (61.6)	49 (57.0)	102 (59.3)
Double therapy	21 (24.4)	27 (31.4)	48 (27.9)
Triple therapy	65 (75.6)	59 (68.6)	124 (72.1)
6-Minute walk distance — m	270.3±104.8	270.7±99.9	270.5±102.1
NT-proBNP — pg/ml	3603.1±4101.2	2687.3±2771.2	3145.2±3519.8
Mean pulmonary artery pressure — mm Hg	57.0±13.4	55.2±12.1	56.1±12.8
Pulmonary vascular resistance — dyn·sec·cm^{-5}	883.2±410.9	874.7±344.2	879.0±378.2
Cardiac index — liters/min/m^2	2.6±0.6	2.6±0.8	2.6±0.7
Pulmonary artery wedge pressure — mm Hg	10.0±3.3	9.8±3.1	9.9±3.2
Hemoglobin — g/dl	12.9±1.9	12.9±1.9	12.9±1.9
Estimated glomerular filtration rate — ml/min/1.73 m^2	65.1±24.6	73.5±29.7	69.3±27.5

* Plus–minus values are means ±SD. Percentages may not total 100 because of rounding. All the measurements summarized in the table were obtained during the screening visit. NT-proBNP denotes N-terminal pro–B-type natriuretic peptide.

† Race was reported by the patient.

‡ This category includes patients who reported their race as Black, Asian, American Indian or Alaska Native, or other.

§ The body-mass index is the weight in kilograms divided by the square of the height in meters.

¶ The length of time since diagnosis of pulmonary arterial hypertension was calculated by adding the number of days from the date of diagnosis to the date of informed consent (enrollment) plus 1 day, then dividing by 365.24.

‖ The Registry to Evaluate Early and Long-Term Pulmonary Arterial Hypertension Disease Management (REVEAL) Lite 2 risk score assesses a patient's 1-year risk of death on the basis of six noninvasive clinical variables: World Health Organization (WHO) functional class, systolic blood pressure, heart rate, 6-minute walk distance, NT-proBNP level, and renal function. Scores range from 1 to 14, with a higher score indicating a greater 1-year risk of death.

** WHO functional classes range from I to IV, with higher numbers indicating greater functional limitations.

†† Background therapy was not prespecified in the protocol; rather, patients were treated according to the judgment of their respective physicians and local clinical practice. Treatments included double or triple therapy with combinations of endothelin-receptor antagonists, phosphodiesterase-5 inhibitors, soluble guanylate cyclase stimulators, prostacyclin analogues, and prostacyclin-receptor agonists. Patients who were receiving prostacyclin infusion therapy were also included in one of the other categories of therapy.

‡‡ This therapy includes intravenous epoprostenol and intravenous or subcutaneous treprostinil.

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committee recommended stopping the trial early for efficacy. A final decision to terminate the trial was made on November 25, 2024.

The analysis of the primary efficacy end point focused on the occurrence of death from any cause, lung transplantation, or hospitalization lasting at least 24 hours for worsening pulmonary arterial hypertension as a first event for each patient. In this analysis, 15 patients (17.4%) in the sotatercept group and 47 patients (54.7%) in the placebo group had at least one such event (Table S12); the hazard ratio was 0.24 (95% confidence interval [CI], 0.13 to 0.43; P<0.001 at the interim analysis) (Table 2). The log-rank P value met the prespecified statistical criterion for superiority (one-sided P value, <0.0021) (Fig. 1A). Of the 62 primary end-point events that occurred as first events, 9 were deaths (in 6 of 86 patients [7.0%] in the sotatercept group and in 3 of 86 patients [3.5%] in the placebo group), 2 were lung transplantations (in 1 of 86 patients [1.2%] in both groups), and 51 were hospitalizations lasting at least 24 hours for worsening pulmonary arterial hypertension (in 8 of 86 patients [9.3%]

Table 2. Primary and Secondary Efficacy End Points.*

End Point	Sotatercept (N=86)	Placebo (N=86)
Primary end point		
Composite of death from any cause, lung transplantation, or hospitalization ≥24 hr for worsening pulmonary arterial hypertension, assessed in a time-to-first-event analysis		
Hazard ratio (95% CI)†	0.24 (0.13 to 0.43)‡	—
Secondary end points§		
Overall survival, assessed in a time-to-event analysis		
Hazard ratio (95% CI)†	0.42 (0.17 to 1.07)¶	—
Transplantation-free survival, assessed in a time-to-event analysis		
Hazard ratio (95% CI)†	0.34 (0.15 to 0.78)	—
Death from any cause		
Patients who died — no./total no. (%)	7/86 (8.1)	13/86 (15.1)
Between-group percentage-point difference (95% CI)	−7.3 (−17.7 to 2.4)	—
Change from baseline in REVEAL Lite 2 risk score		
Median change estimate (range) from baseline at wk 24‖	−3.0 (−3 to −2)	0.0 (0.0 to 0.0)
Hodges–Lehmann location shift from placebo estimate (95% CI)**††	−3.1 (−4.3 to −1.9)	—
Patients with a low or intermediate REVEAL Lite 2 risk score (≤7) at wk 24 — no./total no. (%)††	34/69 (49.3)	11/72 (15.3)
Between-group percentage-point difference (95% CI)	33.1 (18.4 to 47.0)	—
NT-proBNP — pg/ml		
Median change estimate (range) from baseline at wk 24‖	−1233.0 (−1233 to −1233)	255.4 (211 to 263)
Hodges–Lehmann location shift from placebo estimate (95% CI)**‡‡	−2339.1 (−3378.7 to −1299.4)	—
Mean pulmonary artery pressure — mm Hg		
Median change estimate (range) from baseline at wk 24‖	−13.6 (−14 to −13)	5.5 (5 to 6)
Hodges–Lehmann location shift from placebo estimate (95% CI)**‡‡	−21.2 (−27.8 to −14.6)	—
Pulmonary vascular resistance — dyn·sec·cm^{-5}		
Median change estimate (range) from baseline at wk 24‖	−156.6 (−160 to −152)	46.6 (36 to 104)
Hodges–Lehmann location shift from placebo estimate (95% CI)**‡‡	−339.6 (−511.1 to −168.1)	—
WHO functional class		
Patients with improvement — no./total no. (%)§§	48/86 (55.8)	24/86 (27.9)
Between-group percentage-point difference (95% CI)	27.4 (12.9 to 41.0)	—

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Table 2. (Continued.)

End Point	Sotatercept (N=86)	Placebo (N=86)
6-Minute walk distance — m		
Median change estimate (range) from baseline at wk 24‖	45.4 (45.0 to 46.0)	–5.4 (–9.5 to –1.0)
Hodges–Lehmann location shift from placebo estimate (95% CI)**‡‡	63.0 (23.2 to 102.7)	—
Cardiac output — liters/min		
Median change estimate (range) from baseline at wk 24‖	–0.1 (–0.1 to –0.1)	–0.4 (–0.4 to –0.4)
Hodges–Lehmann location shift from placebo estimate (95% CI)**‡‡	0.5 (–0.2 to 1.2)	—
EQ-5D-5L index score¶¶		
Median change estimate (range) from baseline at wk 24‖‖	0.060 (–1.020 to 0.512)	0.007 (–0.358 to 0.740)

* Except where noted otherwise, all analyses were performed in the intention-to-treat population. Protocol-specified multiple imputation methods were used for handling missing data, as described in the statistical analysis plan. Confidence intervals have not been adjusted for multiple comparisons and should not be used to infer definitive treatment effects. For all the analyses of change from baseline, baseline was defined as the last measurement obtained before the first dose of sotatercept or placebo.

† The hazard ratio (sotatercept vs. placebo) was derived from a Cox proportional-hazards regression model with trial group as the covariate and stratification according to the following randomization factors: REVEAL Lite 2 risk score (9 or 10 vs. ≥11) and pulmonary arterial hypertension subtype (associated with connective-tissue disease vs. not associated with connective-tissue disease) with the use of Efron's method for handling ties. Data from patients who had been stratified incorrectly at the time of randomization were analyzed according to the stratum to which they were supposed to have been randomly assigned for all analyses. The analysis was based on the first event that occurred in each patient before the data-cutoff date (July 26, 2024). All the deaths that occurred before the cutoff date were included, regardless of whether they occurred during or after participation in the ZENITH trial, except for deaths that occurred after lung transplantation or after roll over to the SOTERIA study.

‡ P<0.001 (one-sided) for the comparison of the groups at the interim analysis on the basis of the log-rank test, stratified according to the following randomization factors: REVEAL Lite 2 risk score (9 or 10 vs. ≥11) and pulmonary arterial hypertension subtype (associated with connective-tissue disease vs. not associated with connective-tissue disease).

§ A gatekeeping method was used to control the type I error rate across the secondary end points. Secondary end points were tested in a prespecified sequential order, wherein each subsequent end point was to be tested only if the previous end point showed statistical superiority; a one-sided alpha level of 0.0021, or equivalently, a two-sided alpha level of 0.0042, was used. For the secondary efficacy end points that were ineligible for statistical testing, unadjusted 95% confidence intervals are provided. These 95% confidence intervals should not be interpreted as definitive evidence of an effect in the absence of statistical testing.

¶ The P value did not meet the statistical criterion of superiority for sotatercept over placebo.

‖ Shown is the average of the medians across the imputed datasets (with 95% confidence interval) if missing data were imputed.

** The Hodges–Lehmann location shift from placebo estimate is the median of all paired differences.

†† A total of 17 patients in the sotatercept group and 14 patients in the placebo group were excluded from the analysis because they had had a REVEAL Lite 2 risk score of 7 or lower at baseline or they had undergone randomization less than 24 weeks before the data-cutoff date.

‡‡ For between-group comparisons associated with week 24, only patients who had undergone randomization more than 24 weeks before the data-cutoff date were included in the analysis.

§§ This end point was assessed at the end of the double-blind placebo-controlled treatment period. The last measurement obtained before the end of the patient's participation in the trial or the data-cutoff date, whichever came first, was used in the analysis.

¶¶ The EuroQol 5-Dimension 5-Level (EQ-5D-5L) questionnaire index score is the total score of five dimensions of health-related quality of life, weighted according to country-specific or region-specific preferences; scores range from less than 0 to 1, with higher scores indicating better health-related quality of life.

‖‖ The values for observed median change from baseline are shown. Missing data were not imputed.

in the sotatercept group and in 43 of 86 patients [50.0%] in the placebo group).

After the occurrence of a first event that met the definition of the primary composite end point, 16 subsequent events occurred: 5 lung transplantations (all in the placebo group) and 11 deaths (10 in the placebo group and 1 in the sotatercept group). With respect to all the events (first and subsequent), the results for all three components of the composite end point appeared to be consistent with those of the primary end point (Table 3).

The results of a subgroup analysis of the primary composite end point are provided in Figure S3. Findings across all the prespecified subgroups were consistent with the overall treatment effect.

SECONDARY EFFICACY END POINTS

At the time of the data-cutoff date, 7 patients (8.1%) in the sotatercept group and 13 patients (15.1%) in the placebo group had died (Fig. 1B and Table S13); the hazard ratio was 0.42 (95% CI, 0.17 to 1.07) (Fig. 1B and Table 2). The results for overall survival did not meet the criterion for significance at the interim analysis (required P value,

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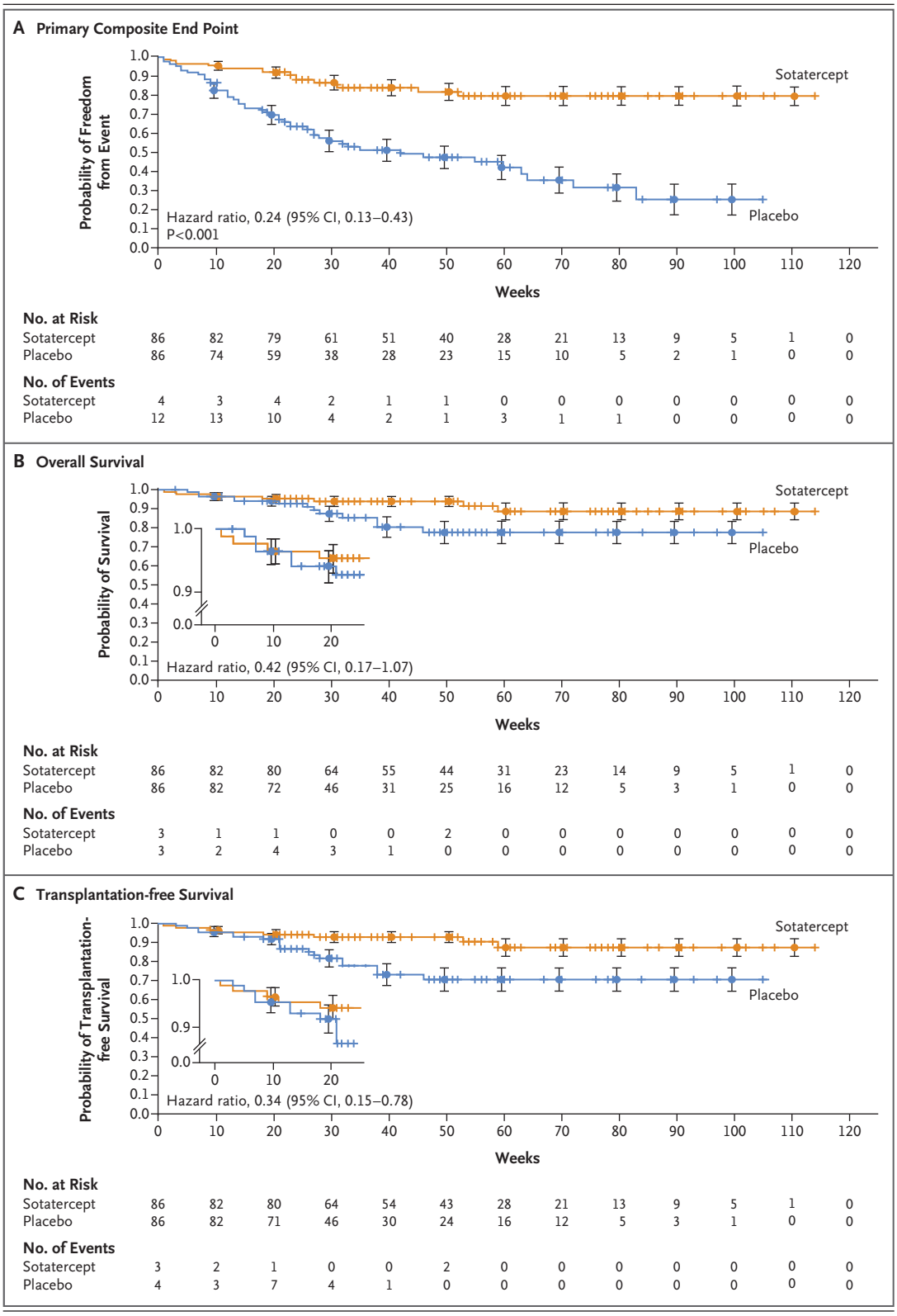

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Figure 1 (facing page). Primary Efficacy End Point, Overall Survival, and Transplantation-free Survival.

Panel A shows the probability of freedom from a first event of death from any cause, lung transplantation, or hospitalization lasting at least 24 hours for worsening pulmonary arterial hypertension. Panel B shows overall survival (defined as the probability of freedom from death from any cause). Panel C shows transplantation-free survival (defined as the probability of freedom from lung transplantation or death from any cause as a first event). The insets in Panels B and C show early event distribution and curve separation on an enlarged y axis. The 95% confidence intervals (indicated by I bars) should not be interpreted as definitive evidence of an effect in the absence of statistical testing. The analyses were performed in the intention-to-treat population, which included all the patients who had undergone randomization, irrespective of whether they received trial medication. The log-rank test was used for comparing sotatercept with placebo, with stratification according to the REVEAL Lite 2 risk score (9 or 10 vs. ≥11) and pulmonary arterial hypertension subtype (associated with connective-tissue disease vs. not associated with connective-tissue disease) in accordance with the randomization schema. Tick marks on each curve indicate censored data. No formal comparisons between the trial groups were made for the individual components of the primary composite end point.

Table 3. Components of the Primary End Point.*

Variable	Sotatercept (N=86)	Placebo (N=86)	Hazard Ratio (95% CI)
No. (%) with ≥1 event	15 (17.4)	47 (54.7)	0.24 (0.13 to 0.43)†
Components of the primary end point — no. (%)‡			
Death from any cause§	7 (8.1)	13 (15.1)	—
Lung transplantation	1 (1.2)	6 (7.0)	—
Hospitalization ≥24 hr for worsening pulmonary arterial hypertension	8 (9.3)	43 (50.0)	—

* This analysis was performed in the intention-to-treat population.
† P<0.001. The primary composite end-point analysis included the occurrence of any first event up to the data-cutoff date (July 26, 2024). All the deaths that occurred before the cutoff date were included, regardless of whether they occurred during or after the patient's participation in the ZENITH trial, except for deaths that occurred after lung transplantation or after roll over to the SOTERIA study.
‡ Each component of the composite primary end point is shown as a standalone end point. A patient is included under more than one component if multiple events meeting the definition of the primary end point were observed.
§ Death from any cause included all deaths that occurred up to the cutoff date, except for those that occurred after lung transplantation or after roll over to the SOTERIA study.

<0.0021); therefore, subsequent secondary end points were not tested.

One patient (1.2%) in the sotatercept group and 6 patients (7.0%) in the placebo group underwent lung transplantation (Table S15). The hazard ratio for transplantation-free survival was 0.34 (95% CI, 0.15 to 0.78) (Fig. 1C and Table 2).

Other secondary end-point results are summarized in Table 2. The results of sensitivity analyses of overall survival and continuous secondary end points are provided in Tables S14 and S11, respectively.

SAFETY

The median duration of follow-up was longer in the sotatercept group than in the placebo group (10.6 months vs. 7.1 months). Adverse events were reported in 98.8% of the patients in the sotatercept group and in 96.5% of those in the placebo group (Table 4). Among the adverse events with an incidence of at least 10% in either group, the events with an incidence of at least 5 percentage points higher with sotatercept than with placebo were epistaxis (in 44.2% vs. 9.3% of the patients), telangiectasia (in 25.6% vs. 3.5%), gingival bleeding (in 10.5% vs. 2.3%), vomiting (in 12.8% vs. 5.8%), and back pain (in 10.5% vs. 4.7%) (Table 4 and Table S16). Discontinuation of the trial medication because of adverse events occurred only in the placebo group (in 4.7% of the patients) (Table 4 and Table S17).

Adverse events that were considered to be related to sotatercept or placebo are listed in Table S18. These events were more common with sotatercept than with placebo (in 65.1% vs. 32.6% of the patients) (Table 4).

Serious adverse events occurred in 53.5% of the patients in the sotatercept group and in 64.0% of those in the placebo group (Table 4 and Table S19). Adverse events that were rated by the investigators as severe were reported in 37.2% of the patients in the sotatercept group and in 50.0% of those in the placebo group (Table S20). No patients in the sotatercept group discontinued treatment owing to serious adverse events (Table 4). Adverse events that led to death occurred in 5 patients (5.8%) in the sotatercept group and in 12 patients (14.0%) in the placebo group. The causes of death are listed in Table S21.

Adverse events of interest that occurred more

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Table 4. Adverse Events.*

Variable	Sotatercept (N=86)	Placebo (N=86)	Point Estimate (95% CI)†
	number (percent)		*percentage points*
Any adverse event	85 (98.8)	83 (96.5)	2.3 (–3.2 to 8.8)
Adverse event leading to discontinuation of sotatercept or placebo	0	4 (4.7)	–4.7 (–11.4 to –0.3)
Adverse event related to sotatercept or placebo‡	56 (65.1)	28 (32.6)	32.6 (17.8 to 45.9)
Adverse event related to sotatercept or placebo leading to discontinuation of sotatercept or placebo‡	0	1 (1.2)	–1.2 (–6.3 to 3.2)
Adverse event leading to death§	5 (5.8)	12 (14.0)	–8.1 (–17.8 to 0.9)
Adverse event related to sotatercept or placebo leading to death‡	0	0	0.0 (–4.3 to 4.3)
Serious adverse event¶	46 (53.5)	55 (64.0)	–10.5 (–24.8 to 4.3)
Serious adverse event leading to discontinuation of sotatercept or placebo¶	0	4 (4.7)	–4.7 (–11.4 to –0.3)
Serious adverse event related to sotatercept or placebo‡¶	3 (3.5)	2 (2.3)	1.2 (–5.0 to 7.8)
Serious adverse event related to sotatercept or placebo leading to discontinuation of sotatercept or placebo‡¶	0	1 (1.2)	–1.2 (–6.3 to 3.2)
Adverse event of interest or special interest‖			
Any such event	74 (86.0)	59 (68.6)	17.4 (5.0 to 29.7)
Bleeding event	54 (62.8)	30 (34.9)	27.9 (13.1 to 41.5)
Cardiac event	13 (15.1)	26 (30.2)	–15.1 (–27.5 to –2.6)
Increased hemoglobin level	11 (12.8)	1 (1.2)	11.6 (4.8 to 20.5)
Telangiectasia	22 (25.6)	3 (3.5)	22.1 (12.4 to 32.7)
Adverse event with an incidence of ≥10% in either or both groups			
Thrombocytopenia**	11 (12.8)	7 (8.1)	4.7 (–4.8 to 14.5)
Cardiac failure	6 (7.0)	10 (11.6)	–4.7 (–14.1 to 4.4)
Right ventricular failure	3 (3.5)	12 (14.0)	–10.5 (–19.9 to –2.2)
Diarrhea	19 (22.1)	15 (17.4)	4.7 (–7.4 to 16.7)
Nausea	16 (18.6)	13 (15.1)	3.5 (–7.9 to 15.0)
Vomiting	11 (12.8)	5 (5.8)	7.0 (–1.9 to 16.5)
Gingival bleeding	9 (10.5)	2 (2.3)	8.1 (0.9 to 16.7)
Fatigue	12 (14.0)	14 (16.3)	–2.3 (–13.4 to 8.7)
Peripheral edema	12 (14.0)	17 (19.8)	–5.8 (–17.3 to 5.6)
Covid-19	14 (16.3)	14 (16.3)	0.0 (–11.3 to 11.3)
Nasopharyngitis	9 (10.5)	11 (12.8)	–2.3 (–12.4 to 7.7)
Pneumonia	9 (10.5)	5 (5.8)	4.7 (–3.9 to 13.7)
Hypokalemia	15 (17.4)	13 (15.1)	2.3 (–9.0 to 13.7)
Back pain	9 (10.5)	4 (4.7)	5.8 (–2.4 to 14.7)
Headache	20 (23.3)	20 (23.3)	0.0 (–12.7 to 12.7)
Dizziness	9 (10.5)	8 (9.3)	1.2 (–8.3 to 10.7)
Epistaxis	38 (44.2)	8 (9.3)	34.9 (22.4 to 46.7)

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Table 4. (Continued.)

Variable	Sotatercept (N=86)	Placebo (N=86)	Point Estimate (95% CI)†
	number (percent)		*percentage points*
Dyspnea	10 (11.6)	21 (24.4)	−12.8 (−24.4 to −1.3)
Pulmonary arterial hypertension††	4 (4.7)	25 (29.1)	−24.4 (−35.4 to −14.0)
Telangiectasia	22 (25.6)	3 (3.5)	22.1 (12.4 to 32.7)
Hypotension	12 (14.0)	9 (10.5)	3.5 (−6.6 to 13.8)

* Adverse events that occurred up to and including day 56 after the last dose of sotatercept or placebo were included in the analysis. This analysis was performed in the safety population, which included all the patients who had undergone randomization and had received at least one dose of sotatercept or placebo, and was performed according to the trial medication administered. Covid-19 denotes coronavirus disease 2019.

† Shown is the point estimate for the between-group difference. The 95% confidence intervals were calculated with the use of the Miettinen and Nurminen method. The confidence intervals should not be interpreted as definitive evidence of an effect in the absence of statistical testing.

‡ These adverse events were suspected to be related to sotatercept or placebo by the trial investigator.

§ The number of patients who died from an adverse event differs from the number of deaths assessed for the primary and secondary end points that included death from any cause as a component. Adverse events leading to death were evaluated in the safety population, with data collected from the first dose of sotatercept or placebo to 56 days after the last dose.

¶ A serious adverse event was defined as any untoward medical event that results in death, is life-threatening, warrants hospitalization or causes prolongation of existing hospitalization, results in persistent or significant disability or incapacity, may have caused a congenital abnormality or birth defect, or warrants intervention to prevent permanent impairment or damage.

‖ Adverse events of interest or special interest are medical concepts that were searched with the use of prespecified standardized *Medical Dictionary for Regulatory Activities* (MedDRA) queries or groupings of preferred terms. Preferred terms are standardized terms used to describe specific adverse events. Adverse events of interest and special interest were selected on the basis of the mechanism of action of sotatercept, preclinical data, or previous clinical findings. Only the adverse events of interest and special interest in which the point estimate for the between-group difference excluded zero are shown in this table. Complete results and details regarding the prespecified search strategies for adverse events of interest and special interest are provided in the Supplementary Appendix.

** These results show the number of patients in whom thrombocytopenia was reported as an adverse event as opposed to the term thrombocytopenia that was used as a prespecified search term.

†† This term refers to the worsening of pulmonary arterial hypertension.

frequently in the sotatercept group than in the placebo group included telangiectasia (in 25.6% vs. 3.5% of the patients; all were reported as cutaneous), increased hemoglobin level (in 12.8% vs. 1.2%), and thrombocytopenia (in 14.0% vs. 8.1%) (Table 4 and Table S22). The imbalance in bleeding events (62.8% with sotatercept vs. 34.9% with placebo) was mostly driven by the occurrence of epistaxis and gingival bleeding. The percentage of patients who had serious bleeding events was balanced between the groups (5.8% in the sotatercept group and 4.7% in the placebo group) (Table S23).

DISCUSSION

In the ZENITH trial, the efficacy and safety of sotatercept was evaluated in patients at high risk for death (WHO functional class III or IV and a REVEAL Lite 2 risk score ≥9), despite receiving double or triple therapy for pulmonary arterial hypertension. The interim analysis showed a 76% lower risk of a composite of death from any cause, lung transplantation, or hospitalization lasting at least 24 hours for worsening pulmonary arterial hypertension with sotatercept than with placebo. On the basis of this result, the data monitoring committee recommended stopping the trial early for efficacy, a highly unusual occurrence for a randomized, placebo-controlled trial involving patients with pulmonary arterial hypertension.

The incremental efficacy of the addition of sotatercept to background pulmonary arterial hypertension therapy probably relates to the mechanism of action of sotatercept.[10-13] By targeting the activin-signaling pathway, sotatercept improves the balance between growth-promoting factors and growth-inhibiting factors, addressing the core pathobiologic features of pulmonary arte-

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rial hypertension.[11,12] The efficacy of sotatercept in this heavily treated population with advanced pulmonary arterial hypertension reflects its role in reducing the risk of major outcomes. The ZENITH trial used a composite end point that incorporated only major outcomes, distinguishing it from previous trials that included less severe outcomes in their composite end points, such as deterioration in exercise capacity, worsening of WHO functional class, and initiation of rescue therapy for pulmonary arterial hypertension.[4-6,16] With respect to subsequent events beyond first events of hospitalization, all three event types of the composite primary end point appeared to occur less frequently among patients who received sotatercept than among those who received placebo. These findings underscore the value of sotatercept as an add-on therapy and its mechanistic ability to induce reductions in the risk of major outcomes, even in a heavily treated population.

The early termination of the trial at the interim analysis due to efficacy had important implications. It led to a loss of clinical equipoise, making continuation of the administration of placebo unethical. In addition, it curtailed the number of accrued events, thereby reducing the statistical power to test secondary end points, including overall survival and transplantation-free survival. Changes from baseline in the REVEAL Lite 2 risk score, the mean pulmonary artery pressure, pulmonary vascular resistance, WHO functional class, the 6-minute walk distance, and the NT-proBNP level were observed, with the magnitude for some of the variables appearing to exceed those seen in the STELLAR trial.[8] This may be attributed to greater disease severity in the ZENITH population and the trajectory of disease progression with receipt of placebo. As was seen in previous phase 2 and phase 3 trials of sotatercept,[8,9,16-18] the reduction from baseline in mean pulmonary artery pressure led to a decrease in pulmonary vascular resistance, whereas cardiac output did not change substantially after treatment with sotatercept.

The safety profile was generally consistent with that seen in previous trials.[8,9] As in the STELLAR trial,[8] the percentages of patients who had serious or severe adverse events, adverse events that led to discontinuation of sotatercept or placebo, and adverse events that led to death were lower in the sotatercept group than in the placebo group. These findings are noteworthy given the longer duration of exposure in the sotatercept group, which would have been expected to lead to a higher incidence of adverse events in that group.

Adverse events associated with sotatercept included increased hemoglobin level, bleeding (mostly nonserious epistaxis and gingival bleeding), telangiectasia, and thrombocytopenia. The percentage of patients in whom serious bleeding events were reported was similar in the two groups. The safety profile reflects off-target effects of sotatercept on erythropoiesis, hemostasis, and microvascular structure. Although thrombocytopenia was observed, it did not account for the imbalance between the groups in the incidence of bleeding events. The ongoing open-label SOTERIA study will help characterize the long-term safety profile of sotatercept.[19]

This trial has limitations. First, the sample size was smaller than that of previous outcome trials in pulmonary arterial hypertension; however, this limitation was partly offset by the number of events accrued and the observed treatment effect. Second, early termination of the trial curtailed the duration of follow-up, which precluded a longer-term assessment of safety and efficacy. The median duration of follow-up was 8.9 months (range, 0.3 to 26.1), constraining the reporting and interpretation of further events including deaths and lung transplantation. Furthermore, hospitalization for worsening pulmonary arterial hypertension was the most common component of the primary composite end point, a finding that should be considered when interpreting the results. Third, the ZENITH trial was enriched for patients with pulmonary arterial hypertension at high risk for death (WHO functional class III or IV and a REVEAL Lite 2 risk score ≥9), which strengthened the applicability of findings to advanced pulmonary arterial hypertension; however, this approach may limit generalizability to patients with less severe disease. Fourth, early discontinuations in the placebo group led to more missing data in that group than in the sotatercept group and an imbalance in the duration of follow-up between the groups. Fifth, changes from baseline in some continuous end points, including

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the 6-minute walk distance, were sensitive to the handling of missing data owing to the incidence of death, lung transplantation, or hospitalization lasting at least 24 hours for worsening pulmonary arterial hypertension.

In this trial, treatment with sotatercept resulted in a significantly lower risk of a composite of death from any cause, lung transplantation, or hospitalization (≥24 hours) for worsening pulmonary arterial hypertension than placebo in this population of patients with advanced pulmonary arterial hypertension who were at high risk for death despite receiving the maximum tolerated dose of background therapy. The trial was stopped early for efficacy at an interim analysis. The safety profile was generally consistent with that seen in previous trials.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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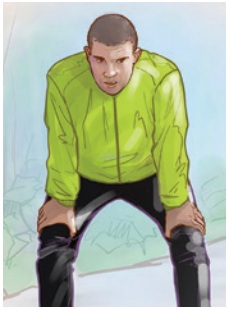
Double Take Video: **A Burning Question**



This video explores the differential diagnosis for progressive dyspnea and cough and highlights the importance of taking a thorough history.

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